

SBC 201 Chapter 31 Special Construction — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–10 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 31, source file Reference\SBC 201 2024\source_reference\Chapter_31 – SPECIAL CONSTRUCTION.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **121** independently checkable numeric records (**60** Verified, **61** Verify source). Unresolved OCR is listed in the register and is not a design-release value.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, structural design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapters 7, 10, 11, 15, 16, 24, 26 or 32; Sections 504.3, 506, 705, 707, 716 or 903.3.1.1; Table 705.5; SBC 401, 501, 701, 801, 901 or 1401; ASCE 55; TIA-222; NFPA 70; ISO 668 / 1496-1 / 6346; CST-GUIDE-101.3-24; commentary examples; or the existing chapter summary. Where Chapter 31 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection is not selected. Walkway travel-distance and several 3104.5 options cite **Section 903.3.1.1** only. NFPA 13 versus 13R is not assumed.
4. Pedestrian links, amenity pools, rooftop or BIPV solar, entry canopies/marquees, terrace/atrium membranes, temporary event structures, rooftop telecom and vehicular gates are unconfirmed.
5. Greenhouse, relocatable-building, flood-hazard public-restroom and intermodal-container programs are not assumed. Those clauses are inventoried for coverage and omitted from lead tables.
6. Section 3114 appears in the extract only as “Not applicable to KSA”; no 3114 code values are adopted.
7. 3104.8 total-width text and Table 3115.8.5.3 are OCR-unresolved. Affected inventory records are **Verify source**.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 31 points to another section/code/standard, or OCR/AHJ must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 31 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 31 application	Required project action
Pedestrian walkway or tunnel	Unconfirmed link between towers, podium buildings or parking	3104 geometry, 2-hour junctions, open-side options and travel distance apply only if a walkway or tunnel is used as a means of travel	Freeze whether any bridge, podium link or tunneled walkway exists; pick one 3104.5 connection option before detailing the junction
Sprinkler standard	Unconfirmed; 903.3.1.1 is the only walkway sprinkler citation	3104.5.2, 3104.5.3 Exception, 3104.5.4 and 3104.9 Exceptions 1 and 3 require a 903.3.1.1 system. A 13R-only tower does not unlock those branches	Do not claim the 75 m / 120 m or 16.5 m / fifth-story walkway options unless 903.3.1.1 is locked
Amenity pool, spa or hot tub	Unconfirmed	3109.1 sends design and construction to SBC 1401 ; an air-supported cover also triggers 3102.8.3 regardless of occupant load	Confirm pool/spa program; extract enclosure and barrier values from SBC 1401, not from this chapter
Rooftop or BIPV solar	Unconfirmed	3111 charges wind to 1609 , roof live load to 1607.14.4 , fire class to 1505.8–1505.9 , BIPV to 1507 , and pathways to SBC 801 Section 1205	Show PV/BIPV on the roof plan before importing those outbound sections
Entry canopy, awning or marquee	Unconfirmed	3105 covering tests and 3106 thickness/projection rules apply only where those elements exist	Schedule canopy/awning/marquee on elevations; keep Chapter 32 encroachment values outbound
Membrane on terrace, atrium or roof	Unconfirmed	3102 applies for 180 days or longer, and for any duration when erected on a building, balcony or deck. Mixed-construction membranes have 6 m / 6.1 m height locks	Flag tensile halls, pool covers and atrium membranes; do not treat a deck-mounted tent as SBC 801-only
Temporary event or construction structure	Unconfirmed; duration of the structure, not seasonal use	3103 applies below 180 days ; permit at > 11 m² with 10 or more persons; travel 30 m or less	Log any pavilion, tent or sales structure and its erection period; structures left up 180 days or more leave this section
Rooftop telecom / broadcast	Unconfirmed	3108 charges TIA-222 , deletes TIA seismic exceptions, and sets escarpment extent at 16 times height	Structural/telecom consultant to confirm mast or guyed tower; keep TIA values outbound

Decision / gap	Current project basis	Why it controls Chapter 31 application	Required project action
Automatic vehicular gates	Unconfirmed site access	3110 charges ASTM F2200 and UL 325	Identify automated vehicle gates on the site plan; pedestrian gates are outside this section
Greenhouse / relocatable / container	Not assumed	3112–3115 and Table 3115.8.5.3 are not typical of this R-2 tower	Re-open those sections only if the program adds a greenhouse, modular building or shipping-container structure
NOC / fire strategy	Unconfirmed	Walkway ratings, membrane inflation, PV firefighter pathways and pool enclosures cannot be concluded from this extract	Engage the qualified local/fire consultants before design freeze

4. Chapter scope

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3101.1	Chapter special-construction scope	This chapter governs membrane structures, temporary structures, pedestrian walkways and tunnels, automatic vehicular gates, awnings and canopies, marquees, signs, towers, antennas, relocatable buildings, swimming-pool enclosures and safety devices, solar energy systems, public-use restrooms on publicly owned flood-hazard land, and intermodal shipping containers	Any listed special-construction element on the project	None stated	Direct	Route each listed element through its Chapter 31 section; do not treat this chapter as a substitute for Chapters 7–12 or 16	Verified

5. Membrane structures

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3102.1	Permanent membrane duration	Air-supported, air-inflated, membrane-covered cable, membrane-covered frame and tensile membrane structures erected for 180 days or longer shall comply with 3102.1 through 3102.8. Shorter independent membranes shall comply with SBC 801 . Membrane structures on a building, balcony, deck or other structure shall comply with this section for any duration	Membrane structure proposed on or beside the building	Non-occupied covers of water/sewage plants, greenhouses and similar facilities need only 3102.3.1 and 3102.7	Conditional	Treat a terrace, podium or roof membrane as a 3102 structure even if it is temporary; do not import SBC 801 tent values here	Verified
3102.2	Tensile and air-supported standard	Tensile membrane structures and air-supported structures, permanent and temporary, shall be designed and constructed in accordance with ASCE 55 . Sections 3102.3 through 3102.6 also apply	Tensile or air-supported membrane	No ASCE 55 stresses or fabric values are stated in this chapter	External verification	Name ASCE 55 on the membrane structural drawing; do not copy fabric prestress or cable sizes into this matrix	Verified
3102.3	Membrane construction type	Noncombustible membrane structures, and noncombustible frame or cable structures with an approved 3102.3.1 membrane, are Type IIB . Heavy-timber frame with an approved membrane is Type IV-HT . Other membrane structures are Type V	Stand-alone membrane building	Exception: plastic less than 9.1 m above any floor in non-public greenhouses, and aquaculture pond covers, need not meet NFPA 701 Test Method 1 or 2	Conditional	Classify a stand-alone tensile hall as IIB, IV-HT or V before applying 3102.4–3102.5; do not use the greenhouse height exception on this tower	Verified
3102.3.1	Membrane and liner material	Membranes and interior liners shall be noncombustible per Section 703.3 or shall meet NFPA 701 Test Method 1 or 2, as appropriate, and the manufacturer's test protocol	Membrane or interior liner	Exception: plastic less than 0.5 mm thick in non-public greenhouses, and aquaculture pond covers, need not meet NFPA 701	Conditional	Specify noncombustible or NFPA 701-listed fabric; keep 703.3 criteria outbound; do not use the 0.5 mm greenhouse exception	Verified
3102.4–3102.5	Membrane area, height and stories	Floor area shall not exceed Section 506 . Membrane structures shall not exceed one story and shall not exceed the height in metres in Section 504.3	Stand-alone membrane structure	Exception: noncombustible membrane structures serving as roofs only are not limited by the one-story / 504.3 cap	External verification	Take area and height from 506 / 504.3 after the 3102.3 type is locked; keep a roof-only noncombustible membrane on the mixed-construction path	Verified
3102.6	Mixed-construction membrane	Membrane used as a portion of a building of another construction type is permitted. Height and area limits shall be those of the host building's type and occupancy	Membrane attached to or forming part of a rigid building	3102.3 through 3102.5 apply instead where the whole structure is membrane or membrane on frame/cable only	Conditional	Size a podium or atrium membrane to the tower's height/area, not as a separate IIB/V hall	Verified
3102.6.1	Noncombustible roof or skylight height	Noncombustible membrane permitted as roof or skylight of any building or atrium of any construction type where the membrane is not less than 6 m above any floor, balcony or gallery	Noncombustible membrane roof or atrium skylight	None stated	Conditional	Set atrium or amenity-roof membrane soffit $\geq$ 6 m above the highest occupied floor, balcony or gallery it covers	Verified
3102.6.1.1	Flame-resistant roof or skylight height	An NFPA 701 Test Method 1 or 2 membrane may be used as roof or skylight on Type IIB, III, IV-HT and V where the membrane is not less than 6.1 m above any floor, balcony or gallery	Flame-resistant membrane on those construction types	This path is not stated for Type I	Conditional	Use 6.1 m only if the host construction is IIB/III/IV-HT/V; a Type I tower roof membrane follows 3102.6.1 noncombustible 6 m instead	Verified
3102.7–3102.7.1	Membrane loads and frame restraint	Structure shall sustain dead loads, tension or inflation loads, and live loads including wind, flood and seismic, in accordance with Chapter 16 . For membrane-covered frames, the membrane shall not be considered to provide lateral restraint of the frame members	Membrane structure, including mixed construction	No Chapter 16 load figures are stated here	External verification	Design the frame without fabric bracing; take environmental loads from Chapter 16	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3102.8.1.1–3102.8.2	Auxiliary inflation and standby power	Air-supported and air-inflated buildings larger than 140 m² shall have an auxiliary inflation system that starts automatically on loss of internal pressure or primary-blower failure. Where auxiliary inflation is required, approved standby power shall start and transfer all required functions at full power within 60 seconds of normal-service failure and shall operate independently for not less than 4 hours	Air-supported or air-inflated structure	Primary inflation with automatic pressure control and over-pressurization protection is required for all such structures (3102.8–3102.8.1). Blowers: continuous-rated motors, inlet screens and guards, weather housing, backdraft dampers, approved uncontaminated inlets (3102.8.1.2)	Conditional	If an air hall exceeds 140 m² , show primary plus automatic auxiliary blowers and a 4-hour standby generator with 60-second transfer on the MEP/life-safety drawings	Verified
3102.8.3	Deflation support clearance	Standby support required after deflation where occupant load is 50 or more, or where the structure covers a swimming pool at any occupant load. Type I roof membranes shall be held not less than 6.1 m above floor or seating. Other membranes shall be held not less than 2.1 m above the floor, seating area or water surface	Air-supported or air-inflated structure with OL ≥ 50 or covering a pool	None stated	Conditional	Provide a standby cable/frame that keeps a Type I roof membrane ≥ 6.1 m and any other membrane ≥ 2.1 m after deflation; pool covers need this support even with few occupants	Verified

6. Temporary structures

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3103.1	Temporary-structure duration	Sections 3103.1 through 3103.4 apply to structures erected for less than 180 days . Special-event structures, tents, umbrella structures and other membranes in that period shall also comply with SBC 801 . Longer erections shall comply with the applicable permanent-structure sections	Structure erected for a limited period	Duration is of the structure, not seasonal occupancy of a permanent shell. A membrane on a building still follows 3102.1 for any duration	Conditional	Log erection and removal dates; if the pavilion remains 180 days or more, design it as a permanent building instead	Verified
3103.1.1	Temporary health and safety conformance	Temporary structures and uses shall conform to this code's structural strength, fire safety, means of egress, accessibility, light, ventilation and sanitary requirements as necessary to protect public health, safety and general welfare	Temporary structure or use	Strength, fire, egress, accessibility and sanitation values are not restated here	External verification	Apply Chapters 7–12, 16 and any cited SBC 501/701 rules to the temporary use; do not waive them because the structure is short-term	Verified
3103.1.2–3103.2	Temporary permit and documents	Permit required for temporary structures covering greater than 11 m² , including connecting spaces with a common egress or entrance, that are used or intended for gathering 10 or more persons. Each installation needs a permit application, construction documents, a site plan, means of egress and occupant load	Temporary structure above the area and gathering thresholds	Smaller structures used by fewer than 10 persons are outside this permit sentence	Conditional	Pull a permit and issue a site/egress plan before erecting any event tent or sales structure over 11 m² intended for 10 or more people	Verified
3103.3	Temporary location and wall rating	Temporary structures shall be located in accordance with Table 705.5 based on the fire-resistance rating of the exterior walls for the proposed type of construction	Temporary structure on the site	No Table 705.5 distances are stated in this chapter	External verification	Set fire-separation distance and exterior-wall rating from Table 705.5; do not invent setbacks here	Verified
3103.4	Temporary exit-access travel	Temporary structures shall conform to Chapter 10 means of egress and shall have an exit-access travel distance of 30 m or less	Temporary structure	Chapter 10 values other than this 30 m cap are not restated here	Conditional	Draw occupant-load and travel paths so every point in the temporary structure is ≤ 30 m to an exit; do not use ordinary Chapter 10 travel tables to relax this cap	Verified

7. Pedestrian walkways and tunnels

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3104.1–3104.1.1	Walkway area and application	Pedestrian walkways and tunnels used as a means of travel, at, above or below grade, shall not contribute to the building area or the number of stories or height of the connected buildings. Walkways: 3104.2 through 3104.9. Tunnels: 3104.2 and 3104.10	Connection between buildings used for pedestrian travel	None stated	Conditional	Keep a complying link off both buildings' area and storey count; do not apply 3104.3–3104.9 tunnel-only checks to a below-grade tunnel	Verified
3104.2	Connected buildings as separate structures	Buildings connected by pedestrian walkways or tunnels shall be considered separate structures	Walkway or tunnel between buildings	Exception 1: same-lot buildings treated as one building per 503.1.2 . Exception 2: for Chapter 11 Type B unit counts, structurally connected buildings and multi-wing buildings are one structure	Conditional	Keep two towers as separate buildings for height/area unless 503.1.2 is demonstrated; still count Type B units as one structure across the link	Verified
3104.3	Walkway construction type	Pedestrian walkway shall be of noncombustible construction	Pedestrian walkway	Exception 1: combustible construction where the connected buildings are combustible. Exception 2: fire-retardant-treated wood per 603.1 Item 1.3 for the walkway roof where connected buildings are not less than Type I or II	Conditional	Specify a noncombustible link on a Type I/II tower; FRTW is permitted only in the roof and only under Exception 2	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3104.4	Walkway contents	Only materials and decorations approved by the building official shall be located in the pedestrian walkway	Pedestrian walkway	None stated	Conditional	Keep retail kiosks, storage and unapproved finishes out of the link; obtain building-official approval for any remaining contents	Verified
3104.5	Walkway-to-building connection options	Connection shall comply with 3104.5.1, 3104.5.2, 3104.5.3 or 3104.5.4	Pedestrian walkway entering a building	Exception: same-lot buildings considered one building per 503.1.2 need not use these four options	Conditional	Pick one numbered connection option on the life-safety plan; do not mix 2-hour, glass, open-side and >2-hour-wall rules on the same junction	Verified
3104.5.1–3104.5.1.2	Two-hour walkway junction	Separate the walkway from the building interior with not less than 2-hour fire barriers per Section 707 and 3104.5.1.1–3104.5.1.3. Connected exterior walls shall be 2-hour fire-resistance rated for not less than 3 m in every direction around the walkway perimeter. Openings in that rated exterior wall shall have opening protectives with a fire-protection rating of not less than 3/4-hour per Section 716	3104.5.1 fire-barrier option	Supporting construction follows 707.5.1 (3104.5.1.3); those values are not restated here	Conditional	Draw a 2-hour barrier at the building interface, a 3 m 2-hour exterior wrap, and 3/4-hour protectives in that wrap; keep 707/716 details outbound	Verified
3104.5.2–3104.5.2.2	Alternative smoke or glass wall	Separating wall shall resist passage of smoke, or be tempered, wired or laminated glass fully wetted by 903.3.1.1 sprinklers with no head/glass obstructions and gasketed frames that deflect without breaking the glass before sprinkler operation — available where connected buildings are more than 3 m apart, or where the walkway and buildings are sprinklered throughout per 903.3.1.1 , the walkway roof is not more than 16.5 m above grade, and the walkway connects to the fifth or lower story	Walkway using smoke-tight or wetted-glass wall	Open parking garages need not be equipped with the automatic sprinkler system	Conditional	Use this option only if building-to-building distance > 3 m or the 903.3.1.1 / 16.5 m / fifth-story set is true; detail either a smoke-tight wall or fully wetted glass	Verified
3104.5.3	Open-side walkway junction	Intersection walls need not be fire-resistance rated where buildings are more than 3 m apart, both walkway sidewalls are not less than 50 percent open with openings uniformly distributed, the roof is not more than 12 m above grade plane, and the walkway connects only to the third or lower story	Walkway with open sidewalls	Exception: walkway sprinklered per 903.3.1.1 — roof not more than 16.5 m above grade plane, connecting to the fifth or lower story	Conditional	If using open sides, show ≥ 50 percent open both sides, roof ≤ 12 m , and no connection above the third story unless the 903.3.1.1 exception is locked	Verified
3104.5.4	Junction at walls above 2 hours	Intersection walls need not be fire-resistance rated where connected exterior walls required by Section 705 are rated greater than 2 hours , the walkway is sprinklered throughout per 903.3.1.1 , the roof is not more than 16.5 m above grade plane, and the walkway connects to the fifth or lower story	Connected exterior walls rated > 2 hours	Both the sprinkler item and the 16.5 m / fifth-story item must be met	Conditional	Use this option only on elevations that already need > 2-hour 705 walls, and only with a 903.3.1.1 walkway at or below 16.5 m / fifth story	Verified
3104.6	Walkway over public way	Pedestrian walkways over a public way shall comply with Chapter 32	Walkway spanning a public way	No Chapter 32 encroachment values are stated here	External verification	Send public-way projection, vertical clearance and permit geometry to Chapter 32	Verified
3104.7	Required-exit walkway access	Access shall be provided at all times to a pedestrian walkway that serves as a required exit	Walkway used as a required exit	Locking, hardware and horizontal-exit rules are not restated here	Conditional	Keep the link unlocked whenever either connected building is occupied; do not isolate one tower after hours if the walkway is an exit	Verified
3104.8	Walkway minimum width	Unobstructed width of pedestrian walkways shall be not less than 0.9 m	Pedestrian walkway	Egress capacity may still require a greater width under Chapter 10	Conditional	Draw a clear walking width ≥ 0.9 m ; if the link is an exit component, also size it to Chapter 10 without importing those factors here	Verified
3104.8	Walkway maximum width	Total width shall be not greater than the published maximum	Pedestrian walkway	Attached extract reads 9 mm , which is not a usable walkway maximum and is not adopted	Conditional	Hold the maximum-width dimension until a published 3104.8 is checked; do not treat 9 mm as the design limit	Verify source
3104.9	Walkway exit-access travel	Exit-access travel length shall be 61 m or less	Pedestrian walkway	Exception 1: walkway sprinklered throughout per 903.3.1.1 — 75 m or less . Exception 2: both sides not less than 50 percent open — 90 m or less . Exception 3: both sides not less than 50 percent open and sprinklered per 903.3.1.1 — 120 m or less	Conditional	Measure travel inside the walkway to 61 m unless a named 903.3.1.1 or 50-percent-open branch is demonstrated	Verified
3104.10	Tunneled-walkway separation	Separation between the tunneled walkway and the connected building shall be not less than 2-hour fire-resistant construction, with openings protected in accordance with Section 716	Tunneled walkway	No opening-protective hour rating is stated in this section	External verification	Provide a 2-hour tunnel/building interface and take opening protectives from Section 716; do not import commentary hour ratings	Verified

8. Awnings, canopies and marquees

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3105.2	Awning and canopy structure	Awnings and canopies shall resist wind, other lateral loads and live loads per Chapter 16 , with allowance for shape and open construction. Protect structural members against deterioration. Awning frames shall be noncombustible, fire-retardant-treated wood, heavy timber per 2304.11, or 1-hour construction, with combustible or noncombustible covers, and may be fixed, retractable, folding or collapsible	Awning or canopy	Chapter 16 pressures are not stated here	Conditional	Specify a noncombustible, FRTW, heavy-timber or 1-hour frame on the canopy typical; take wind and live load from Chapter 16	Verified
3105.3	Awning and canopy covering	Provide an approved covering that complies with one of: NFPA 701 Test Method 1 or 2; flame-spread index not greater than 25 (ASTM E84 or UL 723); or NFPA 286 with no flame to the ceiling during the 40-kW exposure, no flashover, no flame to the outer extremity, and peak heat-release rate not greater than 800 kW	Awning or canopy covering	Exception: fire-propagation and flame-spread index requirements do not apply to awnings on detached one- and two-family dwellings	Conditional	List NFPA 701, FSI $\leq$ 25 , or a passing NFPA 286 covering on the canopy specification; do not use the detached-dwelling exception on this tower	Verified
3106.2	Marquee vertical thickness	Height or thickness from lowest to highest point shall be not greater than 900 mm where the marquee projects more than two-thirds of the lot-line-to-curb distance, and not greater than 2.7 m where the projection is less than two-thirds of that distance	Marquee	None stated	Conditional	Dimension marquee depth vs curb: $\leq$ 900 mm if projection > two-thirds lot-line-to-curb; $\leq$ 2.7 m if < two-thirds	Verified
3106.3	Marquee roof drainage	Skylight portions shall comply with Chapter 24 . Every marquee roof and skylight shall be sloped to downspouts that conduct drainage so it does not spill over the sidewalk	Marquee	Chapter 24 skylight values are not stated here	Conditional	Slope the marquee to building downspouts; send skylight construction to Chapter 24	Verified
3106.4–3106.5	Marquee location and construction	Locate the marquee so it does not interfere with any exterior standpipe, obstruct stairways or exit discharge, or prevent installation or maintenance of street lighting. Support the marquee entirely from the building, construct it of noncombustible materials, design it per Chapter 16 , and protect structural members against deterioration	Marquee	Chapter 16 loads are not restated here	Conditional	Hang a noncombustible marquee from the facade only; keep it clear of standpipes, exit discharge and street lights	Verified

9. Signs and telecommunication towers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3107.1	Sign design and maintenance	Signs shall be designed, constructed and maintained in accordance with this code	Building or site signs	Structural and public-way rules are in Chapter 16 and Chapter 32, not quantified here	Direct	Put signs on the structural and encroachment drawings; do not invent projection or wind values from this section	Verified
3108.1	Telecom tower design standard	Towers shall be designed and constructed in accordance with TIA-222 . Towers shall be designed for seismic loads; TIA-222 Section 2.7.3 seismic exceptions shall not apply. In TIA-222 Section 2.6.6.2, the horizontal extent of Topographic Category 2 escarpments shall be 16 times the height of the escarpment	Telecommunication or broadcast tower	Exception: single free-standing poles supporting antennas not greater than 23 m from top of pole to grade need not be noncombustible	Conditional	Name TIA-222 with seismic design required and escarpment extent 16h ; combustible poles only if height $\leq$ 23 m	Verified
3108.2	Tower guys and access	Guy wires and accessories shall not cross or encroach on any street or other public space, or over above-ground electric utility lines, or on privately owned property without written owner consent. Provide TIA-222 climbing and working facilities. Limit site access as required by applicable OSHA and EPA regulations	Guyed or accessible tower	TIA-222 climb-facility dimensions are not stated here	External verification	Keep guys inside the leased plot and off streets, public space and overhead lines; send climb-safety geometry to TIA-222	Verified
3116.1	In-building telecommunication circuit	Access-point, junction and pull-box, conduit, duct, tray, pathway and electromechanical elements of a telecommunication circuit are recommended to be designed and constructed as per CST-GUIDE-101.3-24	Telecommunication circuit inside the building	The charging text is recommended, not a numeric limit in this chapter	External verification	Cite CST-GUIDE-101.3-24 on in-building telecom drawings; do not import pathway sizes from that guide into this matrix	Verified

10. Pools, vehicular gates and solar

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3109.1	Pool, spa and hot-tub code	Design and construction of swimming pools, spas and hot tubs shall comply with SBC 1401	Swimming pool, spa or hot tub	No enclosure, barrier or water-depth values are stated in this chapter	External verification	Extract pool barriers, decks and equipment from SBC 1401; do not invent those dimensions here	Verified
3110.2–3110.3	Automatic vehicular gates	Vehicular gates intended for automation shall be designed, constructed and installed to ASTM F2200 . Vehicular gate openers, where provided, shall be listed to UL 325	Automatic vehicular gate	Pedestrian gates are outside this section	Conditional	Specify ASTM F2200 gate construction and a UL 325 opener on the site-access detail	Verified
3111.1.1–3111.1.2	Solar wind and roof live load	Rooftop-mounted PV panel systems and solar thermal collectors shall be designed in accordance with Section 1609 . Roof structures that support solar energy systems shall be designed in accordance with Section 1607.14.4	Rooftop PV or solar-thermal collectors	No wind pressure or live-load figures are stated here	External verification	Send rooftop solar to 1609 and 1607.14.4; do not copy those loads into this matrix	Verified
3111.2–3111.2.1	Solar-thermal listing	Solar thermal systems shall be designed and installed in accordance with this section, SBC 701 , SBC 501 and SBC 801 . Light-transmitting plastic covers: Section 2606.12 . Equipment shall be listed and labeled to ICC 900/SRCC 300 and ICC 901/SRCC 100	Solar thermal system	Outbound plumbing, mechanical and plastic-cover values are not stated here	External verification	List ICC 900/901 on solar-thermal equipment; keep 2606.12 and SBC 501/701/801 values outbound	Verified
3111.3–3111.3.1	PV listing and electrical	Photovoltaic solar energy systems shall be designed and installed in accordance with this section, SBC 801 , NFPA 70 (or its equivalent in SBC 401) and the manufacturer's instructions. Panels and modules: UL 1703 or both UL 61730-1 and UL 61730-2 . Inverters: UL 1741 . Grid-connected inverters shall be listed for utility interaction	Photovoltaic system	Electrical clearances and inverter ratings are not stated here	External verification	Require the named UL listings on the PV specification; keep NFPA 70 / SBC 401 and SBC 801 values outbound	Verified
3111.3.2–3111.3.4	Rooftop PV fire class and access	Rooftop-mounted PV panel systems shall have a fire classification in accordance with Section 1505.9 . BIPV installed as roof coverings: fire classification per 1505.8 and design/installation per Section 1507 . Roof access, pathways and spacing shall be provided in accordance with Section 1205 of SBC 801	Rooftop PV or BIPV roof covering	Class letters, pathway widths and setbacks are not stated here	External verification	Coordinate PV fire class with 1505.8–1505.9 / 1507 and firefighter pathways with SBC 801 Section 1205	Verified
3111.3.5–3111.3.5.1	Ground-mounted PV	Ground-mounted photovoltaic systems shall be designed and installed in accordance with Chapter 16 and SBC 801 . Fire-separation distances shall be those determined by the law	Ground-mounted PV array	No millimetre FSD is stated in this chapter	External verification	Design ground arrays to Chapter 16 and SBC 801, and confirm FSD with the AHJ; do not invent a setback here	Verified

11. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger is confirmed (the amenity, link, canopy, membrane, solar array or gate actually exists).
2. Treat **Verify source** rows as hold points. The 3104.8 maximum-width cell is not a design-release value.
3. Do not import SBC 1401 pool barriers, SBC 801 tent or PV-pathway dimensions, Chapter 10 travel other than the 3103.4 **30 m** cap, Table 705.5 setbacks, Chapter 32 encroachments, TIA-222 climb geometry, ASCE 55 fabric values, or Section 716 opening-protective hours.
4. Do not assume NFPA 13 versus 13R. Walkway options that cite **903.3.1.1** remain closed until that standard is locked.
5. Do not apply greenhouse (3102.3 / 3102.3.1 exceptions, 3112), relocatable (3113), KSA-omitted 3114, or intermodal-container (3115 / Table 3115.8.5.3) numbers to this tower unless the gap register is reopened.
6. Do not treat commentary examples (40 mm water column, 20 mil, 1½-hour tunnel openings, 2045 m²) as code cells.
7. Record walkway-option, sprinkler-standard, pool, PV and membrane decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

12. Coverage summary

Internal inventory of the attached Chapter 31 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 121
- **Verified:** 60
- **Verify source:** 61

Counts by top-level section

Top-level section	Records
3101	0
3102	12

Top-level section	Records
3103	4
3104	25
3105	4
3106	3
3107	0
3108	2
3109	0
3110	0
3111	0
3112	0
3113	0
3114	0
3115	71
3116	0

3114 contains no numbered code in this extract. 3115 includes 11 Verified body/footnote records plus **60** Verify-source cells from flattened Table 3115.8.5.3 (15 visible container designations × length, height, side-wall shear, end-wall shear). Those table cells are not adopted.

Coverage cross-check against SBC 201 Chapter 31 Special Construction (2024)_CS.md was topics-only: 180-day membrane/temporary split; deck-mounted membranes; ASCE 55; IIB / IV-HT / V; 6 m / 6.1 m mixed roofs; inflation **140 m² / 4 hours**; temporary permit **11 m² / 10** persons / **30 m** travel; walkway 2-hour / **3 m** / open-side **50 percent / 12 m** / third story; 3104.8 **0.9 m** min and garbled max; travel **61 / 75 / 90 / 120 m**; awning FSI **25** / NFPA 286 **800 kW**; marquee **900 mm / 2.7 m**; solar pointers; pools → SBC 1401. No CS.md value was copied into a matrix cell.

13. Unresolved-source register

Affected table / clause	Why unverified	Control note
3104.8 total width	Charging text reads “not greater than 9 mm. ” That token is not a usable walkway maximum and is not adopted	Project-use maximum-width row is Verify source . Confirm the published 3104.8 maximum before limiting link width. Minimum 0.9 m remains Verified
Table 3115.8.5.3 and notes a–c grid	Flattened HTML concatenates designations, lengths, heights and kN/m shear cells (13 . 5 m, 12 . 301BB, shared rowspans)	No shear, length or height cell is a design-release value. Container design is omitted from project-use as not typical. Footnote a factor of safety of 5 is inventoried as Verified text only